9-Simple DDA algorithm (Digital Differential Analysis)

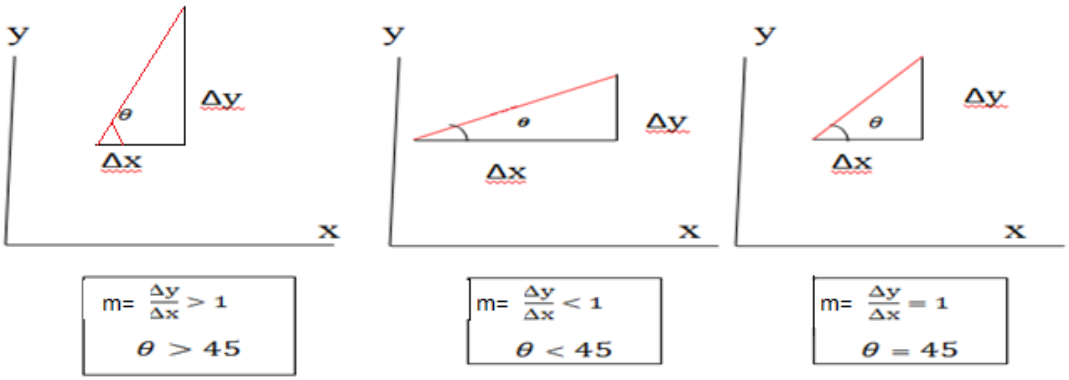
It is a simple scan conversion (The process of representing continuous graphics objects as a collection of discrete pixel called Scan conversion) based on the calculation of

$$\Delta y = m \cdot \Delta x$$

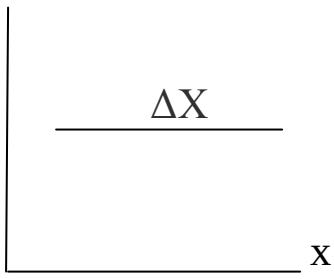
Or

$$\Delta x = \frac{\Delta y}{m}$$

Depending on the line slope. Where $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$



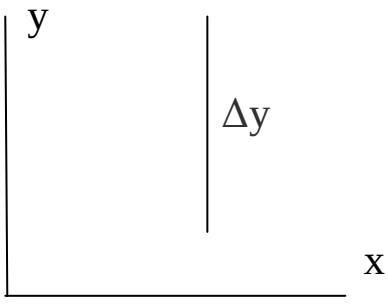
y



Δy=0

$m = \frac{\Delta y}{\Delta x} = \mathbf{0}$

y



Δx=0

$m = \frac{\Delta y}{\Delta x} = \infty$

Procedure-

Given-

- Starting coordinates = (X_1, Y_1)
- Ending coordinates = (X_n, Y_n)

The points generation using DDA Algorithm involves the following steps-

Step-1:

Calculate ΔX, ΔY and M from the given input.

These parameters are calculated as-

- $\Delta X = X_n - X_1$
- $\Delta Y = Y_n - Y_1$
- $M = \Delta Y / \Delta X$

Step-2:

Find the number of steps or points in between the starting and ending coordinates (length).

if (absolute (ΔX) > absolute (ΔY)) ⟶ M< 1

Steps = absolute (ΔX);

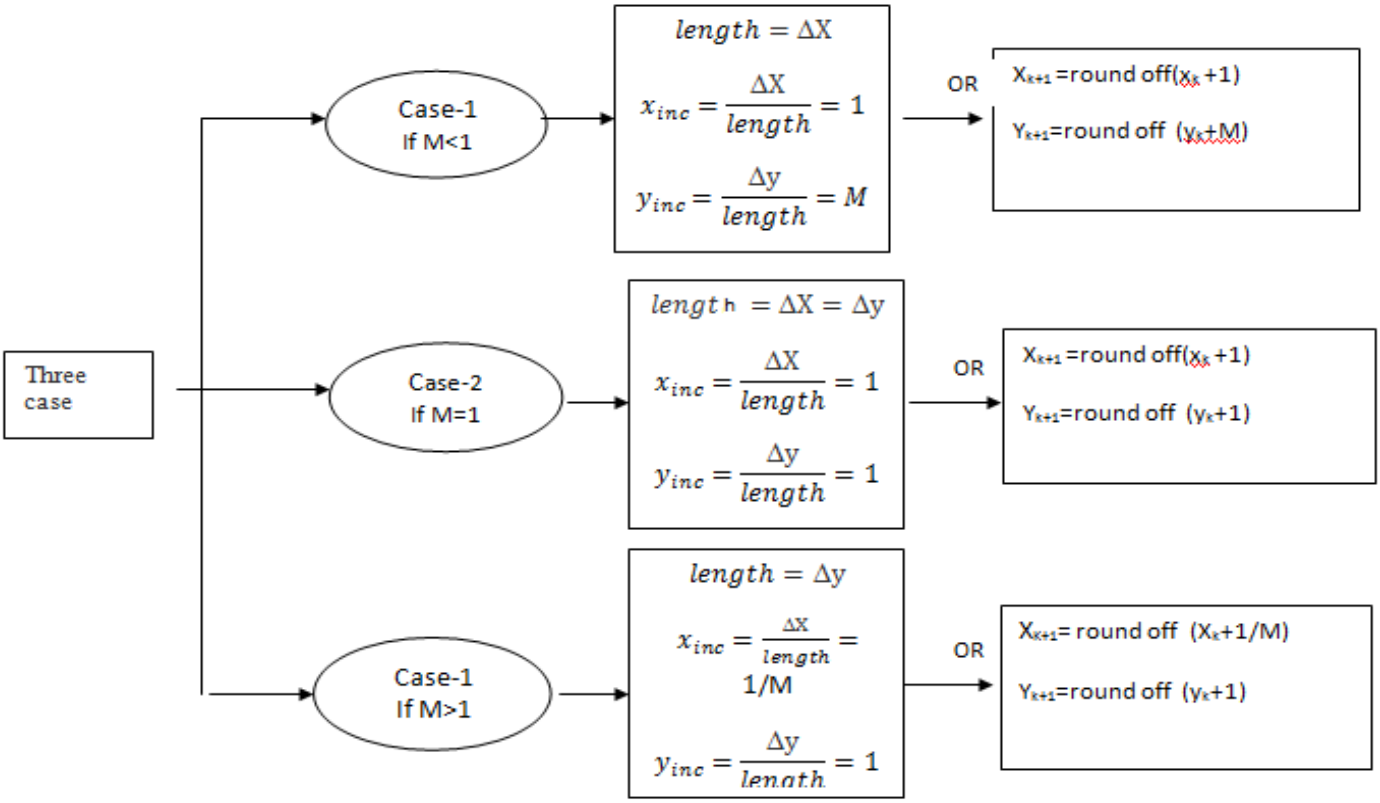
else

Steps = absolute (ΔY); ⟶ M>1

Step-03:

Suppose the current point is (X_k, Y_k) and the next point is (X_{k+1}, Y_{k+1}).

Find the next point by following the below three cases-



❖ This is to draw a line from the left endpoint to the right endpoint. If this processing is reversed from right to left we have:

dx=-1 and $\boldsymbol{y_{k+1} = y_k - m}$

or

dy =-1 and $\boldsymbol{x_{k+1} = x_k - 1/m}$

Step-04:

Keep repeating Step-03 until the end point is reached or the number of generated new points (including the starting and ending points) equals to the steps count.

(At first k=0 and then x or y is increased by 1 from first point to the final point.)

Algorithms of DDA

$dx=x_2-x_1$

$dy=y_2-y_1$

If (abs (dx)>abs(dy))

length=abs(dx)

els

length=abs(dy)

x_{inc}= (dx)/ (length);

y_{inc}= (dy)/ (length);

For(k=0;k<=length ;k++)

{

$x=x+x_{inc;}$

$y=y+y_{inc}$;

Putpixel (x,y);

}

PRACTICE PROBLEMS BASED ON DDA ALGORITHM-

Problem-01:

Calculate the points between the starting point (5, 4) and ending point (12, 7).

Solution-

(x_1,y_1) (x_2,y_2)

(5,4)(12,7)

ΔX=12-5=7

Δy=7-4=3

$m = \frac{\Delta y}{\Delta X} = \frac{3}{7}$ $m < 1$

$|x_2 - x_1| > |y_2 - y_1|$

As m< 1, so case-1 is satisfied. ΔX = $length = 7$

$x_{inc} = \frac{\Delta X}{length} = \frac{7}{7} = 1$

$y_{inc} = \frac{\Delta y}{length} = \frac{3}{7} = 0.4$

k	x	y	X plotted	Y plotted
0	5	4	5	4
1	6	4.4	6	4
2	7	4.8	7	5
3	8	5.2	8	5
4	9	5.6	9	6
5	10	6	10	6
6	11	6.4	11	6
7	12	6.8	12	7

